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A hybrid framework based on CNN-LSTM for Parkinson's disease severity level classification from 3-Axis hand gesture signal under the frequency domain

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A Hybrid Framework Based on CNN-LSTM for Parkinson's Disease Severity Level Classification from 3-Axis Hand Gesture Signal Under the Frequency Domain

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โครงข่ายลูกผสมจากพื้นฐาน CNN-LSTM สำหรับการแบ่งกลุ่มความรุนแรงของโรคพาร์กินสันจากข้อมูลสัญญาณการเคลื่อนไหวมือภายใต้โดเมนความถี่

นายธีรวัต พินิจวัฒนานนท์

วิทยานิพนธ์นี้เป็นส่วนหนึ่งของการศึกษาตามหลักสูตรวิทยาศาสตรมหาบัณฑิต
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Thesis Advisor	Associate Professor Dr. Krerk Piromsopa
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ธีรวัต พินิจวัฒนานนท์ : โครงข่ายลูกผสมจากพื้นฐาน CNN-LSTM สำหรับการแบ่งกลุ่มความรุนแรงของโรคพาร์กินสันจากข้อมูลสัญญาณการเคลื่อนไหวมือภายใต้โดเมนความถี่ (A Hybrid Framework Based on CNN-LSTM for Parkinson's Disease Severity Level Classification from 3-Axis Hand Gesture Signal Under the Frequency Domain) อาจารย์ที่ปรึกษาหลัก: รองศาสตราจารย์ ดร.เกริก ภิรมย์โสภา , อ.ที่ปรึกษาร่วม: ผู้ช่วยศาสตราจารย์ ดร.พัทธมน ปัญญาแก้ว

งานวิจัยนี้นำเสนอแนวทางใหม่ในการสร้างแบบจำลองสำหรับจำแนกความรุนแรงของโรคพาร์กินสันหลายระดับซึ่งถูกออกแบบมาให้สามารถตรวจจับโรคในระยะเริ่มต้นและจำแนกความรุนแรงของโรคได้อย่างเหมาะสม แบบจำลองนี้ใช้ข้อมูลประเภท 6 องศาเสรี (6 Degree of Freedom) ในการเรียนรู้ซึ่งข้อมูลถูกบันทึกและจัดเก็บจากการให้ผู้ป่วยทำแบบทดสอบอย่างง่ายในการกดคีย์บอร์ดเพื่อบันทึกข้อมูล งานวิจัยนี้ได้ทำการวิเคราะห์แบบรูปการเคลื่อนไหวของนิ้วมือผู้ป่วยขณะกดคีย์บอร์ดมาวิเคราะห์ความรุนแรงของโรคเนื่องจากความผิดปกติในการขยับนิ้วมือเป็นตัวชี้วัดที่มีความสัมพันธ์โดยตรงกับความรุนแรงของโรคพาร์กินสัน ข้อมูลที่เก็บมาจากเซ็นเซอร์แบบสวมใส่นั้นจะถูกเตรียมข้อมูลผ่านการคัดเลือกคุณลักษณะและการปรับให้เป็นมาตรฐานก่อนที่จะถูกแปลง ให้อยู่ในรูปของโดเมนความถี่จากการแปลงฟูรีเย (Fourier Transform) หลังจากนั้นข้อมูลความถี่จะถูกกรองผ่านตัวกรองแบบเกาส์เซียน (Gaussian filtering) เพื่อลดสัญญาณรบกวนและเน้นแบบรูปที่เกี่ยวข้องของการขยับนิ้วที่มีความสัมพันธ์กับการจำแนกความรุนแรงของโรค ข้อมูลที่ผ่านการประมวลผลจะถูกใช้ในการสร้างแบบจำลองจากโครงข่ายประสาทเทียมแบบหน่วยความจำระยะยาว-ระยะสั้น (Long Short-Term Memory) ประกอบกับการป้อนข้อมูลโครงข่ายย่อย (subnet feeding) เพื่อใช้ในการจำแนกความรุนแรงของโรค ผลลัพธ์จะถูกประด้วย เม-ทริกซ์ความสับสน (confusion metrics) บนชุดข้อมูลทดสอบทดสอบที่แบ่งออกมาจากชุดข้อมูลหลัก 1,553 รายการ แบบจำลองที่ได้มีความแม่นยำของการจำแนกกลุ่มแบบหลายระดับอยู่ที่ 74% โดยมีความไว (Sensitivity) ในการจำแนกผู้ป่วยระยะเริ่มต้นอยู่ที่ 90% และมีอัตราส่วนลบเท็จ (False Negative Rate) อยู่ที่ 9.7% งานวิจัยนี้แสดงให้เห็นถึงถึงศักยภาพของการผสานเทคโนโลยีเซ็นเซอร์แบบสวมใส่เข้ากับการวิเคราะห์ที่อิงตามโครงข่ายประสาทเทียมแบบหน่วยความจำระยะยาว-ระยะสั้น ซึ่งสามารถนำไปสู่การพัฒนาการตรวจโรคที่เข้าถึงได้ง่ายขึ้น ลดการรุกรานของโรค และปรับเปลี่ยนได้ตามความต้องการ สำหรับการคัดกรองและประเมินความรุนแรงของโรคพาร์กินสันในระยะเริ่มต้น ซึ่งท้ายที่สุดแล้วจะช่วยสนับสนุนการตัดสินใจทางการแพทย์ที่ดีขึ้นได้

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Theerawat Pinijwattananon : A Hybrid Framework Based on CNN-LSTM for Parkinson's Disease Severity Level Classification from 3-Axis Hand Gesture Signal Under the Frequency Domain Advisor: Associate Professor Dr. Krerk Piromsopa , Co-advisor: Assistant Professor Dr. Pattamon Panyakaew

This study introduces a novel multi-class classification model designed for the early-stage detection and severity assessment of Parkinson's Disease (PD). Machine learning model uses data from a 6 Degrees of Freedom (6DoF) sensor, collected during a simple keyboard-tapping test. Our research substantially analyzes finger movement patterns, since movement dysfunction like finger dexterity reduction is a primitive PD indication. First, raw data from wearable sensors are preprocessed by feature selection and normalization. Then, the data were converted into the frequency domain using Fourier transformation. Eventually, Gaussian filtering is applied to reduce noise and highlights relevant pattern. These processed spectral features are used to train a Long Short-Term Memory (LSTM) network with subnet feeding for PD classification. Result were evaluated by confusion metrics on a test set split from dataset of 1,553 records. Our model accomplished a 74% accuracy, with 90% sensitivity for detecting early-stage PD (symptom level 1), and a 9.7% false-negative rate. This work highlights how combining wearable sensor technology with LSTM-based analysis can lead to an accessible, non-invasive, and scalable solution for early PD screening and severity assessment, ultimately supporting better clinical decisions.

Field of Study	Computer Science	Student's Signature ....
Academic Year	2024	Advisor's Signature ....
		Co-advisor's Signature ....
		Co-advisor's Signature ....

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LIST OF ABBREVIATIONS

5-HT	Serotonin
6DoF	6 Degree of Freedom
Ach	Acetylcholine
CNN	Convolutional Neural Network
DA	Dopamine
DFT	Discrete Fourier Transform
FFT	Fast Fourier Transform
FSR	Force-Sensitive Resistors
GABA/Ca2+	Gamma Amino Butyric Acid/Calcium
IMU	Inertial Measurement Unit
LSTM	Long Short-Term Memory
PD	Parkinson's Disease
PSD	Power Spectral Density
ReLU	Rectified Liner Unit
RNN	Recurrent Neural Network
SNpc	Substantia nigra pars compacta

Chapter 1 Introduction

This chapter aims to provide an overview of Parkinson's Disease (PD) severity levels, standard diagnostic protocols, and potential bottlenecks in the diagnostic process. We begin by detailing the symptoms of PD progression across different stages, specifically levels 1, 2, 3, 4, and 5. The clinical protocol is explained, covering both standard procedures and advanced methods for screening and diagnosing PD. Moreover, specific diagnostic issues will be highlighted and applicability of the machine learning-based solutions proposed in this work. This chapter also outlines the objectives, scope of work, and anticipated contributions of the research.

1.1 Problem Statement

Parkinson's disease (PD) is a common and progressive neurodegenerative disorder, primarily affecting the central nervous system and the cause still unknown. PD is characterized by the gradual onset of both motor and non-motor dysfunctions, which advancing impair the patient's quality of life. The clinical progression of PD is commonly assessed by clinical procedure and grouping based on the Hoehn and Yahr scale [4] [19], a standard tool to classify the severity of symptoms into five distinct stages.

In the early stage, classified as level 1, symptoms are typically mild and may include tremors, slight rigidity, or bradykinesia (slowness of movement), usually affecting only one side of the body. At this point, patients often maintain normal daily functioning with minimal disruption. However, as the disease advances to level 2, symptoms become bilateral, involving both sides of the body, although balance remains largely unaffected.

By stage 3, the progression of PD becomes more pronounced. Patients may begin to experience postural instability, slower movements, and difficulties with coordination. While they may still retain some independence, daily activities become increasingly challenging. Importantly, the psychological impact of the disease also becomes more noticeable at this stage, with many individuals reporting depression, anxiety, sleep disturbances, and a reduced overall sense of well-being due to persistent stiffness and chronic pain.

In stage 4, motor symptoms significantly limit mobility. Although patients may still be able to walk or stand unassisted, their movements are severely impaired, and they generally require continuous support for daily tasks. Medical supervision becomes essential to manage symptoms and prevent complications.

Stage 5 represents the most advanced phase of PD. Patients in this stage are typically unable to walk or stand without help and are often confined to a bed or wheelchair. Full-time assistance is required for all activities of daily living. At this point, the disease profoundly affects not only physical abilities but also cognitive and emotional health, necessitating comprehensive, multidisciplinary care [13] [19].

Table 1 Hoehn and Yahr scale

Stage	Description
1	Unilateral involvement only usually with minimal or no functional disability.
2	Bilateral or midline involvement without impairment of balance.
3	Bilateral disease: mild to moderate disability with impaired postural reflexes; physically independent.
4	Severely disabling disease; still able to walk or stand unassisted.
5	Confinement to bed or wheelchair unless aided.

While early-stage Parkinson's Disease (PD) symptoms often cause only minor disruptions, their subtle and gradual onset makes early detection particularly challenging [4], [14], [21]. However, early and accurate diagnosis is crucial because timely intervention can help slow disease progression and significantly improve long-term patient outcomes. Traditional PD evaluation primarily relies on clinical assessments of motor function. These assessments include tasks like the finger-tapping test to gauge finger dexterity [14], along with evaluations of reflexes, muscle rigidity, and postural stability [4]. Patients typically undergo a range of neurological examination tasks, and their performance is then assessed against standardized diagnostic criteria.

However, these clinical evaluation methods are inherently subjective and susceptible to inter-rater variability, which can lead to inconsistencies in diagnosis, particularly during the early stages of PD when symptoms are mild and less clearly observable. Although advanced neuroimaging techniques and biomarker analyses offer greater diagnostic accuracy, their acquisition remains limited due to high costs and restricted accessibility, especially in resource-limited or underserved healthcare environments.

The escalating prevalence of PD underscores the urgent need for precise, accessible, and cost-effective diagnostic tools for early detection and severity evaluation. Wearable sensor technology, when integrated with machine learning, presents a promising solution to this challenge. These sensors possess the capability to capture intricate motor abnormalities, including variations in gait, tremor frequency, and finger dexterity, with exceptional accuracy and non-invasiveness. Subsequently, machine learning models are trained on the meticulously collected sensor data to discern underlying patterns and establish predictive frameworks for PD diagnosis and classification.

1.2 Objectives

- To develop a preprocessing framework for 6DoF sensor signals, transforming them into the frequency domain using the Fourier transform.
- To implement a multi-class classification model capable of reliably and accurately predicting the severity level of PD.
- To enable the model to report PD severity scores with high sensitivity, based on the Hoehn and Yahr scale, ranging from level 1 to level 3.

1.3 Scope of Work

- This Parkinson's disease (PD) classification model utilizes medical records obtained from Chulalongkorn Hospital.
- The model was designed to process data collected from a 6DoF sensor, following the data acquisition protocol established by Chulalongkorn Hospital.
- The model focuses exclusively on classifying the severity stage of PD.

1.4 Contributions

This study introduces a novel method for the early detection and severity classification of Parkinson's disease (PD). The proposed approach utilizes data captured from a 6DoF sensor during a keyboard-tapping task. LSTM-based networks are employed to extract subtle movement patterns in finger dexterity that are often imperceptible through conventional clinical evaluations or visual inspection. The primary contributions of this research are as follows:

- The development of an LSTM-based classification model designed to assess PD severity levels 1 through 3, following with the Hoehn and Yahr standard scale.

- Empirical validation of the model's performance using a dataset comprising 1,553 samples, achieving an overall classification accuracy of 74% and a sensitivity of 90% for early-stage PD (level 1).

- The proposal of a practicable and cost-efficient screening tool for PD, particularly suited for deployment in settings with limited access to specialized diagnostic resources.

1.5 Summary

This chapter illustrates the progression of PD symptoms, from initial muscle rigidity and movement instability to comprehensive body dysfunction. We highlight the significant challenge of early PD detection. Using standard diagnostic methods often challenging due to subtle early symptoms or the high cost of advanced techniques. The benefits of early treatment, facilitating timely intervention for patients, form the fundamental basis for our proposed model.

Chapter 2 Background Knowledge

In this section, the basis knowledge relevant to the proposed work is reviewed, covering both medical fundamentals of PD and deep learning techniques. First, the causes, stages, symptoms, diagnosis, and treatment methods of PD were examined to understand how clinicians manage the disease, as described in [12], [13] and [23]. Then, signal handling methods and amplitude data are presented, focusing on feature selection and feature extraction techniques primarily based on [7]. Finally, the concepts of Fourier transformation and frequency domain analysis are introduced, illustrating how they support efficient time-series and frequency-based analysis, as outlined in [8], [10], [15] and [25].

2.1 Parkinson's Disease

2.1.1 Cause of movement impairment

During the PD development, some symptoms like impairment movement can be initiated alongside. Most painful movements occur from the uncommonly chemical functional inside patient brain, for instance, dopamine (DA), serotonin (5-HT), acetylcholine (Ach), or Gamma Amino Butyric Acid/Calcium (GABA/Ca2+). To clarify, brain area named Substantia nigra pars compacta (SNpc), which secretes DA, is affected from the PD improvement making DA produced unusually. DA is a primarily brain chemical and plays role as an inhibitor neurotransmitter, which regulates the neuron part that controls the body movement balancing. While DA level is irregularly presented, the activity from movement neurons is expressed without DA inhibit-controller, resulting in patients having hard to control their body movement. 5-HT is also involved in the body movement as a controller like DA while Ach is more general, expressed as down regulated in various neurological diseases including PD. In addition, GABA manages the Ca2+ influx inside cell mitochondria. In PA case, Ca2+ buffering system is damaged causing Ca2+ excitotoxicity leading to neuronal loss in the SNpc [13].

2.1.2 Symptoms

2.1.2.1 Early symptoms

Early symptoms are difficult to clarify due to the slowness of development. Symptoms include tremors, posture difficulty, soft speech, slow handwriting, abnormal facial expression, problems with limb movement, loss of focus, fatigue, and depression. These early symptoms can occur on one side of the body and spread through the whole patient's body sooner or later.

2.1.2.2 Primary motor symptoms

Symptoms considered as primary motors are resting tremors, rigidity, slow movement, and balance problems. Resting tremors can be observed via the shaking of body part initiate from one side to whole body. While the resistance of muscle extension during the movement, increasing stiffness and pain, is an indicator of rigidity. Slow movement or medical term as "Bradykinesia" is considered by facing against having non-control movement, lower facial expression, decreasing of extent muscle gesture, or even eye blinking and voice volume are declined significantly. Some can face losing the reflex mechanisms and cause instability while standing.

2.1.2.3 Primary non-motor symptoms

In distinction, non-motor symptoms are much more difficult to observe due to the symptoms indirectly coming from muscle stiffness, but from the brain's chemical instability. For example, depression and insomnia can happen to PD patients while the disease is developed. Moreover, cognitive recognition, which can't complete their sentence due to tongue issue, and dementia, a mental health issue, are also considered as the PD indicators competently.

2.1.2.4 Secondary motor symptoms

Developing from the primary stages, difficulties in swallowing and chewing occurred from the muscle that controls around the mouth has been lost. Around 90% of patients lose their voice control and cannot deliver their messages. The secondary level also affects to sexual function with ejaculation difficulties in male and involuntary urination during sex in female.

2.1.2.5 Secondary non-motor symptoms

Severity is increasing from the primary non-motor symptoms. Patient gestures change and become painful due to the muscle rigidity and abnormal postures. An autonomous nervous system has been interrupted and makes it hard to control the body temperature, causing excessive sweating, oily skin, scalp, or hardy dryness. Also, smooth muscles are affected by PD leading to malfunction in some systems, like urinary problems due to the urinary bladder muscles issue or suffering from cardiovascular disease issue due to the blood artery and vein. Within these symptoms, patients usually face emotional breakdowns leading them to be fearful, insecure, and induce them to avoid socializing and a new environment.

2.1.3 Diagnosis

There are only a few tests that are available for PD diagnosis including diagnosis through biomarker, MRI to examine brain's area, and neurological test to measure the body reflexes. Within limits, it is causing difficulties in PD diagnosis efficiently and accessibility. In addition, PD evaluation needs to be considered based on their medical history and neurological examinations. The standard test for the PD diagnosis is an examination of different reflexes and limb movements, in which an accurate diagnosis requires the presence of at least two of three major symptoms. Measuring the reflexes is looking for bradykinesia, measuring the capability of clamp finger and thumb, tremor index, and muscular rigidity, tested by moving the neck, upper limbs, and lower limbs. PD severity can be classified into stages by following Hoehn and Yahr scale, again as shows in Table 1.

2.1.4 Treatment

2.1.4.1 Drug treatments

Since DA levels play an important role in PD, medicine treatment can be grouped by dopaminergic relation which can be classified into 3 groups. (1) Dopaminergic drugs mainly control the DA level in patient's brain. For example, Levodopa focuses on DA levels restoring, Monoamine oxidase-B inhibitors reduce the DA produced excluding from the main DA reservoir, or Dopamine agonists to increase DA levels in the SNpc. (2) Non-dopaminergic drugs are managed the others brain chemical like Ach. For instance, Ach is the most crucial neurotransmitter in the brain and while DA is low, Ach is induced, then over-excitation happened, ending up with tremor and muscle stiffness.

Anticholinergic drugs are handling the struggles by narrow down the Ach effect. (3) The others are controlling the non-motor symptoms, including depression and anxiety.

2.1.4.2 Surgery treatments

These kinds of treatments mainly emphasis to reduce the motor symptoms, especially during the advanced stages of PD. Doctors can attempt surgical lesions like pallidotomy and thalamotomy to decrease PD suffering in patients carefully.

2.1.5 Progression monitoring

Monitoring PD severity can be done by using advanced techniques, which need high cost to accomplish. Neuroimaging is one of the novel options that can monitor the DA levels in the brain via MRS and MRI. It's helpful for studying the physiological status of tremors, stiffness, and emotional breakdown. The other option is using biomarkers and checking under the radioactive environment, for example, radioactive nucleotides. However, this technique requires high cost and complexity and determination as non-practical for screening in general purpose. Other options like neurological tests can be used, but lacking efficiency and reliability due to the tests are subjective and relied on the therapist's perspective and experience.

2.2 Feature Extraction and Feature Selection

In terms of medical analysis, feature selection and extraction are considered one of the biological processes that improve medical diagnosis. Typically, determined features have an important impact on classification accuracy, time complexity, and the number of examples required for the training step. Medical diagnosis, which usually describes the patterns of illness, is based on symptom observation and diagnostic tests. Both evaluations are linked, and the time spent finalizing the execution is crucial to avoid the bottleneck pathway in the hospital. Additionally, feature selection and extraction require certain criteria to accomplish, including evaluation criteria, dimensionality space, and an optimization procedure.

Features selection and extraction contain various options, with each technique might be well-suitable for some kind of dataset, but nor the others. In this section, researchers tend to display some widely used methods for handling the attributes.

2.2.1 Feature Extraction Method

The curse of dimensionality is the common issue in deep learning, especially in pattern recognition problems like image classification or signal recognition. If sample image, size $n \times n$, has been fed into the convolutional layer for image classification, then kernel needs to run through the image and consequences in spending cost complexity in $O(n^2)$ domain. To avoid unnecessary tasks during the process, raw information needed to be trimmed or extracted before using as a learning dataset. Some feature extraction techniques are described in this section.

2.2.1.1 Non-transformed Signal Characteristics

This transformation is suitable for moments, power, amplitude information, or images preprocessing. A two-dimensional signal, like images or amplitude, has two ways to proceed by a non-transformed technique. (1) Assuming that image presents in a consecutive sequence placing in the one-dimensional and can be handled by one dimensional random process. (2) Considers as two-dimensional and handles the image by two-dimensional random process, called as “random field”.

2.2.1.2 Transformed Signal Characteristics

Some datasets need to transform before performing any further step. The notion of this method is to extract the features from the raw information, which detail density is preserved as much as possible, and the noisy residue is relegated after the transformation. One of the most functional tools to deal with high dimensional data is to apply Principal Component Analysis (PCA) into the data. PCA provides the procedure to reduce the high dimensional data into a small number of uncorrelated variables, which also represented the same feature information but less of dimension. The other useful tool that is excellent for the properties concentration is Discrete Fourier transform (DFT). DFT is applied to the data and tends to derive a feature set base in the coefficients format, which maintains the correlation between datapoint.

2.2.2 Feature Selection Methods

2.2.2.1 Exhaustive Search

The main idea of this search is accomplishment the maximum available features combination to minimize the classification error. Exhaustive search is done by selecting a portion of the best features out of the maximum available features, leading to the total number of feature combinations rising sharply. Huge cost complexity is required as a trade-off to handle the exponentially increased task.

2.2.2.2 Branch and Bound Algorithm

This technique was invented to reduce the high-cost complexity of exhaustive search by determining the optimal feature set without explicit evaluation of all possible feature combinations.

2.2.2.3 Max-min feature selection

Feature selection based on minimal or maximal values that influence the learning process. It aims to reduce inconsistency in data point scaling, which mitigate uncertainty behavior of the dataset when performing model construction.

2.2.2.4 Sequential forward and backward selection

While branch and bound algorithm is created to fix a high-cost complexity issue, however, the algorithm still faces over complexity when apply to the medical information. The solution to the trade-ff problem might come from balancing the trade-off between the optimality and the efficiency which are described as suboptimal solutions. Therefore, many medical images tend to use suboptimal methods which are called "sequential forward selection" (SFS) and "sequential backward selection" (SBS). SFS is a bottom-up technique by filling the empty feature set with individually best measurement. SBS is a top-up technique by eliminating each smallest loss from the whole feature set. The disadvantage of this method is both SFS and SBS rely on nested effect, which eliminated features cannot be reconsidered or removed later respectively.

2.3 Fourier Transform and Frequency Analysis

As signal data, the complex signal waves are assembled from the various spectrum of simple sine waves. By applying the Fourier transform to the signal, the representations can be decomposed, which is called a "prism" effect as illustrated in Figure 1. This prism effect can be calculated by the mathematical procedure which describes below:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-i2\pi vt}dx$$

Where $X(f)$ is the inner product between the signal $x(t)$ and the complex-value sinusoid, while $i = \sqrt{-1}$ and the calculation boundary is infinite. $X(f)$ is representing as a function of frequency which conducted by the Hertz (s^{-1}) unit. The calculation products exhibit in magnitude m (and the phase θ) of the sine wave that closet to the $x(t)$. In addition, Fourier transformation is invertible, which means the time and frequency domain can be converted seamlessly and ensure that the transformation is well-preserved the raw information.

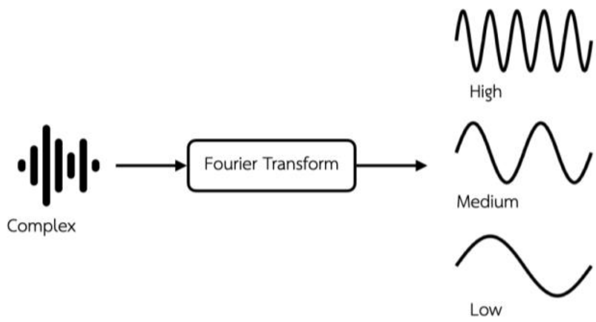

Figure 1 The Fourier transform simplifies the complex signal into a series of simple sine waves, each represent its own frequency.

The Fourier transform provides a well quantitative perspective of the time and frequency domain, especially in modern data analysis. However, the cost for calculating the Fourier integration can be huge and the integral sum needs an equation under the time circumstance from infinite condition, which is not practical for a real-world problem. To shape the problem boundary, data is considered in the time frame from 0 to $N-1$, then the equation becomes discrete Fourier transform (DFT), which is finite and enables calculating in computers directly.

$$X(n) = \sum_{k=0}^{N-1} x[k]e^{-j2\pi nk/N}dx$$

where $$n = 0,1,2,\dots,N-1$$

However, the term of frequency f in the Fourier transform is quite different from the term of frequency f in the DFT. The Fourier transform can take on any value from the integration products, while the frequencies from DFT are all sum and multiple products of the integer within the window over the taken from DFT. Fast Fourier transform (FFT), as same criteria as DFT, requires a compromise: long window provides a well frequency resolution while a short window dispenses a well-locating the event under the time.

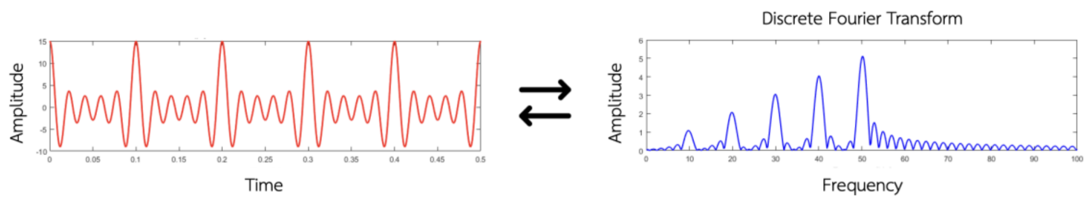

Figure 2 The time domain consists of both short and long amplitudes. The frequency spectrum displays peaks corresponding to individual sine waves. A wave can be converted between the time and frequency domains, as each domain provides a representation of the other.

The frequency domain is exceptionally useful for signal analysis. In this context, the time and frequency domains are two sides of the same coin and can be converted seamlessly between representations. The frequency domain is also capable of capturing patterns that may not be visible in the time domain, making data more suitable for analysis in the frequency domain, as illustrated in Figure 2.

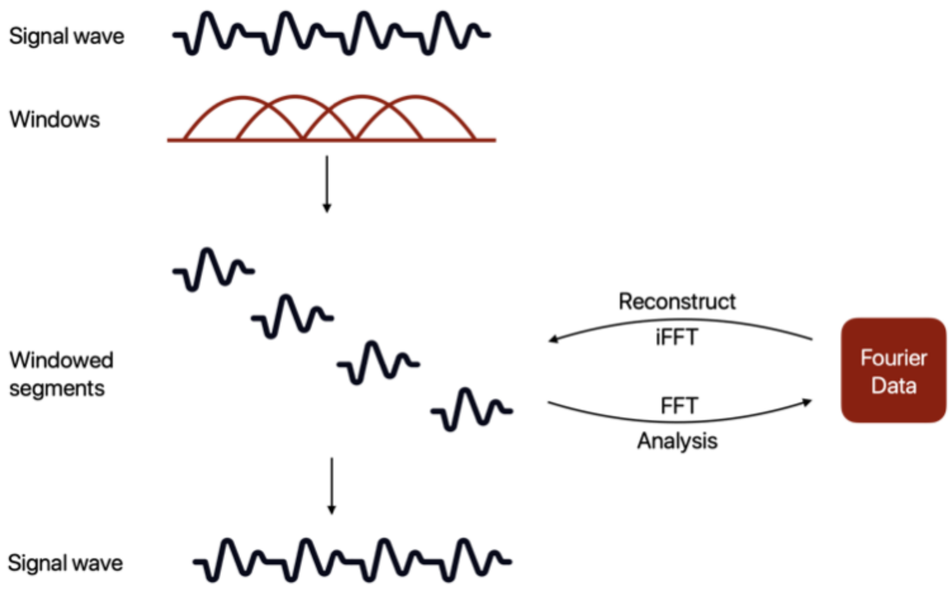

Figure 3 The Short-Time Fourier Transform (STFT) is demonstrated for signal analysis and reconstruction. Windows are defined along the signal, framing it into short segments. These segments are analyzed using the FFT and can also serve as the basis for reconstructing the original signal.

The short-time Fourier transform (STFT) is well-suitable for a long signal when a Fourier window cannot cover all the signal and when the wave signal has a better time localization. The concept of STFT is to use a window function $\omega(t)$ to zeroes all in a short time interval, consequent in all events is localized in that time interval. By the shaped window, the windows are overlapped, shifted, and summed by following the STFT mathematical equation:

$$X_{STFT}(f,t) =< x(t), w(t-\tau)e^{2j\pi ft} >$$

which can be rewritten to a similar DFT as:

$$X_{STFT}[n,i] =< x(k), w(t-\tau)e^{2j\pi nk/N} >$$

where, n denotes the frequency at which the FFT is computed. The Short-Time Fourier Transform (STFT) provides a spectral snapshot that moves along the time axis, resulting in inner products over a series of windowed short segments. These segments can be used to reconstruct the original time-domain signal, meaning that the signal can be resynthesized from its localized frequency components, as illustrated in Figure 3.

2.4 Evaluation Metrics

Hence, clinical models emphasize the importance of not only overall accuracy but also sensitivity and the false negative rate (FNR), as these metrics reflect how effectively clinicians can manage patient care. These evaluation metrics are particularly critical in assessing the performance of patient classification models, as many studies use them as fundamental indicators of model reliability and clinical applicability [3], [11], [14], [16], [17], [20], [21], [22], [24], [26], [28]. The evaluation metrics selected for this study are defined as follows:

- **Accuracy:** the proportion of correctly classified patients to the total number of samples, calculated as:

$$\frac{TP + TN}{TP + TN + FP + FN}$$

- **Sensitivity (Recall):** the ability to correctly identify individuals with Parkinson's disease, calculated as:

$$\frac{TP}{TP + FN}$$

- **False Negative Rate (FNR):** the proportion of patients with Parkinson's disease classify incorrectly as non-PD (healthy), calculated as:

$$\frac{FN}{TP + FN}$$

where TP is true positive, TN is true negative, FP is false positive, and FN is false negative.

Table II presents the macro-sensitivity, which reflects the overall sensitivity of the model by averaging the sensitivity scores across all classes, whereas Table III presents the micro-sensitivity, which provides a more detailed view of sensitivity at the individual class level.

2.5 Summary

In this chapter, we examine the potential causes, symptoms, treatments, and patient tracking methods associated with PD from a medical perspective. A high severity level of PD can significantly disable patients from undertaking their daily lives. While current treatments are limited and primarily serve to maintain the disease's progression, timely intervention may be highly effective only when diagnosis occurs at an early stage.

Several attempts have been made to construct classification models using sensor-based data to assist clinicians in making more accurate and sensitive diagnoses. Standard techniques employed for data preprocessing include feature selection, feature extraction, and relevant motor symptoms, frequency domain analysis. These methods are expected to facilitate the development of a proposed classification model. Additionally, evaluation metrics are explained for validating the classification model. This validation step is crucial for establishing the reliability of the proposed model.

Chapter 3 Literature Review

This chapter categorizes the literature review into three principal sections. The initial section provides an overview of the machine learning models and dataset types widely employed by researchers. Subsequently, the impact of finger dexterity on both PD severity levels and model performance is elucidated. The concluding section investigates published CNN-based classification models from other studies, focusing on those utilizing wearable sensor data for the classification of patients versus healthy individuals.

3.1 Parkinson's Disease: a review of literature

According to the literature review published by [18] in 2021, the overall trends in the application of technology and machine learning in the medical field were highlighted, particularly in the diagnosis of PD. The review noted that PD involves both motor and non-motor symptoms, which are often difficult to distinguish and observe. Over the past decade, machine learning has been applied to various data modalities, including handwriting [16], body movement [5], neuroimaging, voice [26], cardiac scintigraphy [2], serum biomarkers [25], and MRI data [20]. Previous studies revealed that machine learning approaches for PD diagnosis are mainly focused on motor symptom analysis, kinematics, and data from wearable sensors [18]. Among the 448 models reviewed across 209 studies, the most frequently used techniques included support vector machines and their variants (130 studies), neural networks (62 studies), ensemble learning (57 studies), k-nearest neighbors (33 studies), regression methods (31 studies), decision trees (28 studies), Naïve Bayes (26 studies), and discriminant analysis (12 studies).

3.2 Finger Dexterity and Frequency-Based Assessment of PD

Subtle motor impairments, such as reduced finger dexterity, often emerge in the early stages of PD. In response, recent studies have investigated frequency-based metrics as potential indicators for early detection. [14] emphasized the clinical importance of evaluating finger dexterity using frequency features obtained from keystroke-tapping tasks in neurological experiments. The study involved participants of varying ages, genders, PD severity levels, etc., allowing for analysis across a diverse possible parameter to strengthen the validity of the hypothesis. These participants performed a 15-second typing task involving both near tapping (keys 'b' and 'm') and far tapping (keys 'q' and ']') while wearing custom-designed finger-mounted sensor. The analysis focused on key parameters, including typing frequency, typing velocity, cumulative errors, and digraph characteristics.

The study identified strong associations between keystroke dynamics and finger motor function, with significant increases in typing errors and altered typing speeds observed in individuals with Parkinson's Disease (PD). While signs of impaired dexterity may not be immediately apparent, they tend to emerge progressively as muscle activity accumulates. This delayed manifestation, especially a decline in tapping performance, is consistent with the characteristic reduction in movement speed seen in PD patients during motor assessments.

Following statistical analysis of the collected features, the researchers constructed a binary-class decision tree model to differentiate PD patients from healthy individuals. The model achieved a classification accuracy of 70%, demonstrating the potential of keystroke dynamics as a diagnostic tool. However, the scope of the model was limited to distinguishing between PD and non-PD cases, without addressing the more intricate task of classifying PD severity levels. This is particularly relevant for early-stage PD (Hoehn and Yahr stages 1–3), where motor symptoms are often subtle, and more granular classification remains a clinical challenge.

3.3 Neural Network Based Classification Model

To explore the application of signal processing-based model for PD classification, [21] developed a deep learning model leveraging 3-axis wearable sensor data. Tremor signals were recorded from hand movements along the x, y, and z directions, capturing key information related to movement intensity, tremor patterns, and compensatory activity. The unprocessed time-series data were first standardized before feeding as an input of the neural network, composed of seven hidden layers. Rectified Linear Unit (ReLU) activation functions were employed throughout the hidden layers to mitigate the vanishing gradient problem, followed by a flatten layer and a softmax output layer for classification. The model was trained using backpropagation over 30 epochs and validated using the validation set.

The proposed model achieved an accuracy of 92.4%, outperforming baseline models that utilized binary classification techniques (80%) and a CNN-LSTM approach (90%). These results indicate that multi-dimensional sensor data, when processed through a well-structured neural network, can effectively distinguish PD from conditions with similar motor symptoms, such as essential tremor. However, the model was constructed as a binary-class classifier, which may not be ideally suited for identifying early-stage PD, where motor symptoms are more subtle and can overlap with healthy motor variability. In addition, the study primarily emphasized overall accuracy, with limited analysis of other evaluation metrics such as sensitivity, specificity, or precision, which are also critical for clinical applicability.

For outstanding model performance with prerequisite neurological tasks, [22] proposed a multi-class classification model for Parkinson's Disease (PD) using a CNN-LSTM architecture trained on the PhysioNet dataset, which contains vertical ground reaction force (VGRF) signals collected during walking tasks. The VGRF signals were obtained using pressure sensors attached to the soles of the patients ’feet, capturing the force exerted vertically during each step. These measurements reflect detailed gait dynamics, an essential indicator of motor function deterioration in PD patients, making them especially valuable for both diagnosis and severity staging. To enhance signal clarity, the raw VGRF data underwent empirical mode decomposition (EMD), a technique that decomposes the signal into intrinsic mode functions (IMFs) by iteratively subtracting the average of its upper and lower envelopes. These IMFs, which isolate oscillatory components associated with motor control, were then used as input features for the CNN-LSTM model. To improve generalization, the model incorporated L2 regularization to reduce overfitting and used the Adam optimizer for effective parameter tuning.

The model achieved a high classification accuracy of 98.32% and a sensitivity of 98.29%, demonstrating the effectiveness of combining signal processing techniques with deep learning. However, the study acknowledged several limitations. Since VGRF signals are obtained from gait cycles, the approach is inherently limited to patients who are still able to walk, thereby excluding individuals with more advanced stages of PD. Additionally, the CNN-LSTM model imposes a high computational cost due to its complexity and the extensive hyperparameter tuning required for optimization, which may limit its feasibility for broader clinical deployment.

In the study conducted by [24], machine learning techniques are applied to enhance sensor-based activity recognition. The research primarily utilizes accelerometer and gyroscope data, which are commonly employed in this domain. However, the use of frequency-based data augmentation has not been extensively explored. To address this, the study introduces a method where important frequency components of various activities are selectively masked, serving as a comparison group against unmodified amplitude data. Both datasets undergo filtering and ensemble processing to examine the impact of frequency masking. The results demonstrate that the proposed method achieves higher recognition accuracy than the standard ensemble approach. Furthermore, the study shows that recognition performance can be improved by strategically emphasizing or suppressing specific frequency components.

In a related with CNN-LSTM based, [11] employed a CNN-LSTM hybrid model to predict power flow direction. The dataset consisted of two main components: baseline power consumption data and weather information, both structured as amplitude-based time-series data. Prior to model training, the data were normalized to ensure consistency. The study found that the CNN-LSTM model trained solely on baseline data did not yield significantly different performance compared to baseline models. However, when both baseline and weather data were used as inputs, the model demonstrated improved predictive accuracy. The authors concluded that the CNN-LSTM architecture is well-suited for time-series analysis due to its ability to effectively capture both temporal dependencies and spatial patterns inherent in such data.

Machine learning has been applied to a wide range of data types using various classification techniques to categorize PD stages and severity levels. Nevertheless, data acquisition remains a significant challenge, as many methods are not accessible to all PD patients. Studies utilizing sensor-based data, such as in [21] and [22], have successfully distinguished PD from essential tremor and classified PD severity levels by employing advanced data capturing, preprocessing, and CNN-based models. Furthermore, CNN-LSTM models constructed from frequency-domain-processed data have been shown to effectively enhance sensor-based activity recognition. In addition, combining multiple data channels, as demonstrated in [20] research on power consumption and weather information, has proven successful in predictive modeling. Collectively, these findings suggest that signal data from wearable sensors should be transformed into the frequency domain and processed using CNN-LSTM architectures to extract informative patterns. In summary, PD severity stages can be classified by analyzing frequency-based signal plots alongside complementary features.

3.4 Summary

In this chapter, we observed the substantial enthusiasm researchers have demonstrated in developing various diagnostic procedures in the past decade. Some studies reveal a significant correlation between finger dexterity and movement dysfunction, such as a decrease in movement velocity. Others have successfully constructed high-accuracy CNN-based classification models using wearable sensor datasets, which have largely influenced subsequent work in this area. Despite these advancements, some approaches necessitate specific tools or procedures to ensure accurate classification results, which may limit diagnostic accessibility. Furthermore, the performance of CNN-LSTM architectures with different datasets supports their suitability for time-series analysis, owing to their inherent capabilities.

Chapter 4 Methodology

This chapter outlines the methodology employed in this work. We begin by enumerating the data collection process from the hospital and the resulting data structure. The dataset comprises two distinct sections, each preprocessed separately using different techniques described within the chapter. Finally, the analysis techniques, including normalization and frequency domain analysis, are explained.

4.1 Dataset

The collected data consists of two primary components: the six degrees of freedom (6DoF) signal and the record conditions, which include patient information such as age, gender, and PD state. During data collection, two sensors were attached to the index and middle fingers, and patients were instructed to perform a keyboard tapping task designed by [14] to capture finger movement signals. This task involves two types of tapping strokes: a near tapping (on the "B" and "M" keys) and a far tapping (on the "Q" and "]" keys), each performed for 15 seconds. Each patient performs at least three repetitions to minimize variability and account for inconsistencies. A total of 1,553 records were collected, comprising four states: state 0 with 610 records, state 1 with 142 records, state 2 with 528 records, and state 3 with 257 records, forming a comprehensive dataset for model training and evaluation.

To further clarify the collected signal, the data records in 6DoF format. This type of data describes how an object moves and rotates in 3D space, captured through acceleration and gyroscopic measurements. For example, as shown in Figure 4, panel (2) illustrates what the 6DoF data capture, while the time column represents the timestamp of each event within the task period. In the "ay" column demonstrates the vertical movement along the Y-axis. Specifically, at the first timestamp, the finger is located at the 916 y-point; at the second timestamp, the finger moves to the 860 y-point, indicating vertical displacement. This example highlights how the collected signal points can be used to plot and analyze changes in finger movement as the patient performs the tapping task.

(1)

id	patient	hand	num	tapping	gender	age	dominant	stage
5568159	Non PD	Left	0.0	Far	Femal	65	1	2.0
3847256	PD	Left	2.0	Far	Male	72	0	3.0
4358859	Non PD	Left	0.0	Far	Male	57	1	3.0

(2)

time	gx	gy	gz	ax	ay	az	gx2	gy2	gz2	ax2	ay2	az2	key
0.03	-71	255	-19	16292	916	4464	32	4	71	6716	180	21016	0
0.17	-72	-2	-20	16260	860	4472	35	-3	62	6732	192	21156	0
0.20	-69	-1	-19	16525	920	4516	33	-1	63	6720	196	21048	0
0.36	-70	-256	-24	16208	1012	4844	35	-2	57	6704	104	20904	0
0.39	-74	-1	-23	16256	824	4520	33	3	65	6648	144	21236	0

Figure 4 Recorded data from Chulalongkorn Hospital since 2019. The dataset consists of two sections: (1) record conditions, and (2) time-series data from two fingers, represented in 6DoF at each timestamp. The final dataset includes 12 columns corresponding to the 6DoF signals from each finger, along with an additional auto-generated key.

The amplitude plot is used to visualize the collected signal, as shown in Figure 5. Each channel has an individual internal scale; for instance, accelerometer signals range from -25,000 to 25,000, while gyroscope signals range from -10,000 to 10,000. These scale differences can introduce inconsistencies when plotting and may negatively impact model performance. To address this issue, normalization is applied to the data before plotting, ensuring that all signals are scaled between 0 and 1. The amplitude plot illustrates how each finger moves and rotates along different axes. Notably, Channel 2 provides a clearer example: the accelerometer data exhibits a wave pattern that has the potential to reveal or isolate characteristic features in the frequency domain. By extracting these patterns, it becomes possible to classify PD severity levels by analyzing the waveforms generated during this simple neurological task. Since finger dexterity is directly related to finger impairment, this approach offers valuable insights into the severity of PD symptoms.

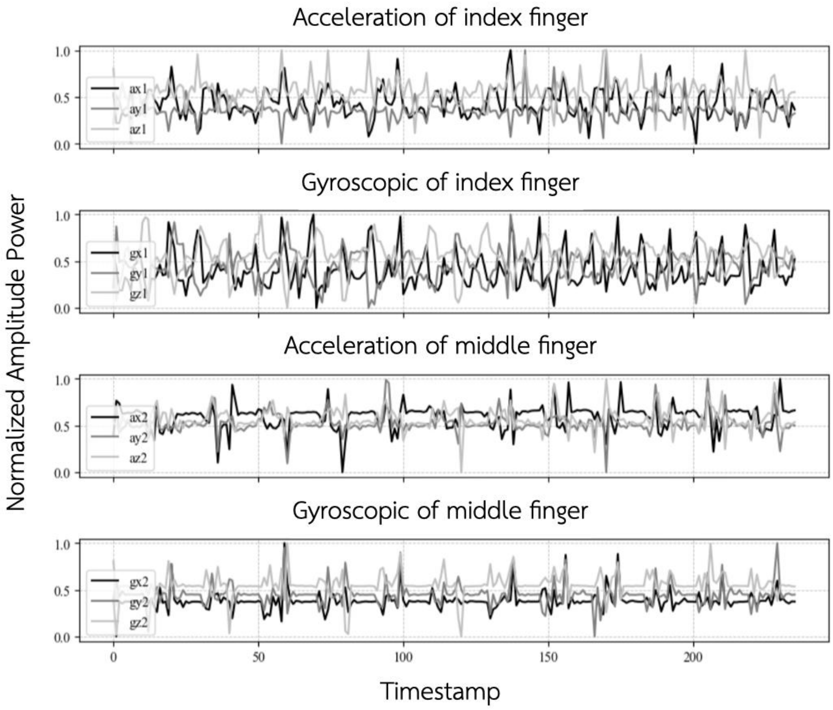

Figure 5 Amplitude plot of the normalized 6DoF signal. Channel 1 represents the signal from the index finger, while Channel 2 represents the signal from the middle finger, respectively.

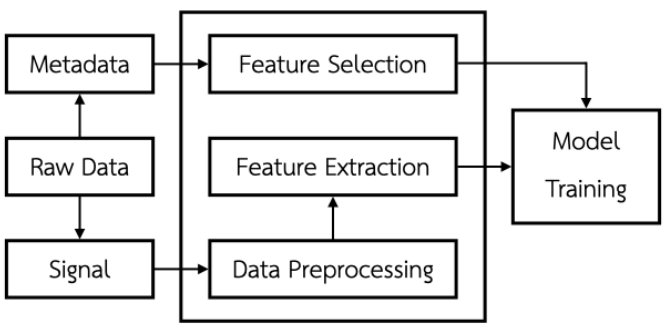

Figure 6 Signal Preprocessing Pipeline

4.2 Methodology

Since the collected data remains in two main streams, they should be handled separately. The record condition section underwent feature selection and then encoded into numeric data for model training compatibility. The 6DoF signal data were transformed and analyzed under the frequency domain before feeding into the input layer of the model. The overall pipeline, as shown in Figure 6, is described in this section below, including data preprocessing and data analysis.

4.2.1 Feature Selection

The record conditions section, illustrated in Figure 4 (1), contains a variety of data collected while patients performed a keystroke-tapping task. Some attributes may help improve model performance, while some may not significantly affect models. Additionally, a high number of attributes can lead to the curse of dimensionality, providing too much information for the model to learn during training and potentially causing overfitting. To address this, we apply feature selection by creating distribution plots for each feature and analyzing their data distribution. Attributes well-suited for model training should display a normal, symmetrical, bell-shaped distribution, whereas asymmetrical or skewed plots indicate data imbalance.

Based on the results, certain asymmetrical plots can be pruned to reduce the impact of imbalanced distributions and to enhance the model's robustness against irrelevant biases [6]. Nevertheless, some features, such as the patient ID, may not exhibit an asymmetrical plot, yet they might not contribute beneficially to the model learning process. These identifiers are unique and not directly related to the disease; their utility lies in patient management and tracking within a hospital setting. Therefore, this attribute should be pruned as it holds no relevance to the disease. Such attributes often necessitate human understanding to facilitate the removal of non-related columns before the collected data are utilized as model training materials.

4.2.2 Data Preprocessing

The 6DoF remains scale inconsistent across the dataset, which may affect weight calculation or tuning progression during the deep learning step. For example, acceleration signals range from -25,000 point to 25,000 point, while gyroscope signal range from -10,000 to 10,000, which could disturb model performance. To mitigate this issue, we apply min-max normalization techniques to shape the data scale between 0 to 1 range. This transformed provide consistency scale across dataset, which ensure the 6DoF signal are suited in the same scale.

Another potential issue appears from dislocations in signal recording periods, often attributable to human error, which lead to variations in timestamps and inconsistent experimental durations. This issue was mitigated by applying a windowing technique, defining the window size to capture a selected signal period for model training. For example, patients may exhibit slight misalignment from the task instruction in a couple seconds at the initiation, however, they typically rectify the task thereafter. This misleading overhead can induce bias learning into the model's performance by incorporating incorrect periods during the training step. This leads to poor performance on unseen datasets, in other words, insufficient generalization for practical usage. From that concern, window size was applied to trim sequences, isolating the tapping periods for feature extraction and improving dataset reliability [6].

4.2.3 Feature Extraction

As previously mentioned by [14], finger dexterity phenotypes correlate with PD severity level. To investigate this, we transformed 6DoF data using Fast Fourier

Transform (FFT) to analyze finger dexterity patterns under the frequency domain. Sensors were attached to both index finger and middle finger, yielding two sets of 6DoF data representing one patient's finger-tapping signal. Given that the acceleration data across the x, y, and z axes share the same scale and origin in 3D space, they were considered as a single dataset before applying FFT to obtain their Fourier representation. This approach resulted in four distinct Fourier representations per patient: one for index finger acceleration, one for index finger gyroscope, one for middle finger acceleration, and one for middle finger gyroscope.

Remarkably, Fourier representation is seamlessly convertible with the frequency domain, hence, the frequency plot can be directly applied to the preprocessed data. However, a noisy baseline may still appear during analysis due to signal saturation and complexity, which could interfere with pattern observation and translation. To mitigate this noise, a Gaussian filter was applied to the Fourier representations, thereby helping to smooth the signal under the frequency domain with reduced background noise and minimal signal distortion [10].

4.2.4 Periodogram

Periodogram is computational tools designed to transform a sample spectrum into a smooth, continuous representation of discrete frequencies. Mathematically, this process estimates the contribution of various frequencies to the overall variance, thereby converting a discrete Fourier representation into a smooth curve plot [1] as shown in Figure 7, adopted from [10]. While Periodogram offer detailed into frequency contributions, their precision doesn't improve with longer data durations. To overcome this limitation, two main strategies are typically used. Averaging neighboring strips: This involves smoothing the Periodogram by combining adjacent frequency bands. Segmenting and averaging: The time series can be divided into shorter segments, with a Periodogram calculated for each. These individual Periodogram is then averaged to produce a more stable result.

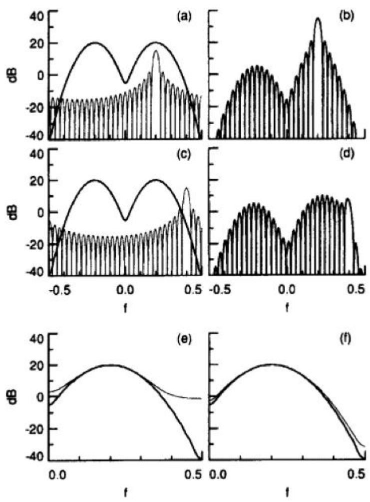

Figure 7 Illustrates the Periodogram methodology, which explains how frequency domain information is abstracted into a smoothened image.

4.2.5 Proposed Architecture Model

In our design, we focus on pattern recognition across different severity levels within the frequency domain, which constitutes a majority component of the classification functional. Additionally, recognizing the potential benefit of a dual baseline approach, an idea inspired by the work of [11], we implemented a double baseline strategy. This strategy utilizes two main data streams, 6DoF signals and record conditions, to enhance our model's classification performance. By applying a subnet technique, our aim is to ensure that the model effectively leverages information from both the signal and the record conditions for optimal overall performance.

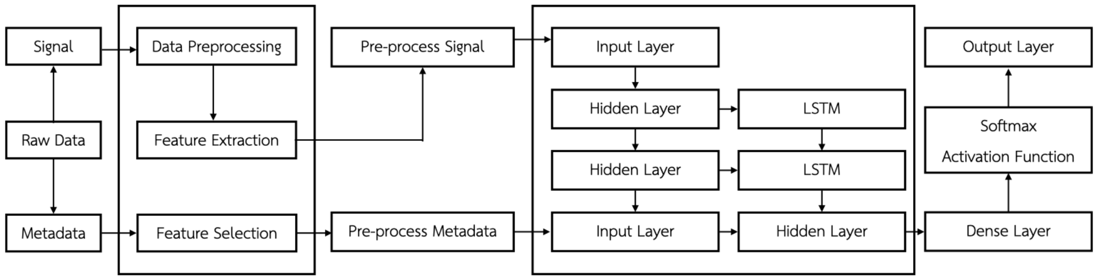

Figure 8 Proposed LSTM-based subnetwork architecture for Parkinson s disease severity rating.

Furthermore, the finger dexterity symptom, as captured by wearable sensors, is relevant to the PD severity level, and early-stage recognition could significantly benefit patient treatment. Hence, our hyper-tuning parameters are specifically focused on the sensitivity of PD level 1. This approach, however, involves a trade-off with other metrics, such as overall accuracy and the accuracy of later-stage classification.

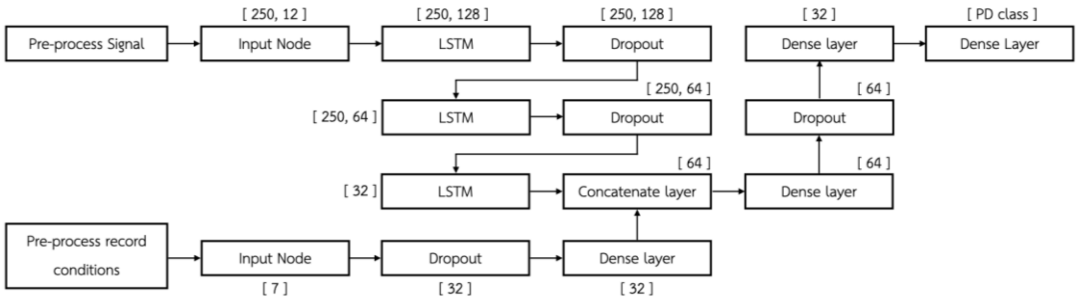

Figure 9 Displays the LSTM-based hidden node architecture that incorporates a subnet aggregate layer.

The proposed architecture is presented in Figure 8. On the left side of the figure, the raw data were separated into two main streams. The first stream, record conditions (metadata), was processed using a feature selection approach, which involves plotting data distributions and validating them to prune non-contributory attributes. The second stream, 6DoF data, encountered data cleaning, windowing selection, and normalization before being transformed into its Fourier representation.

To elaborate on our model's architecture, Figure 9 illustrates the technical details. The first data stream is fed into an input layer consisting of 12 tuples, each containing 250 data points. This input then progresses through three LSTM layers, with each layer connected to a dropout layer set at a 0.3 rate and validated by a ReLU activation function.

Simultaneously, the second data stream is introduced into a separate input layer, comprising 7 columns. This stream bypasses the initial dropout and dense layers and is then concatenated with the output of the LSTM branch, forming a 64-node layer. The combined node subsequently passes through additional dense and dropout layers, before a softmax activation function leads to the final output layer, sized to be compatible with the number of PD classes.

The preprocessed signal from the 6DoF stream serves as the input layer for the model and undergoes training via an LSTM-net, culminating in a LSTM intermediate layer. Concurrently, the record conditions were fed into a separate subnet input layer, as illustrated in Figure 8. After traversing the subnet, this two end were integrated and then passes through the remaining layers. The final output was validated with a classification report and a confusion matrix, respectively. We optimized the model using the Adam optimizer, as suggested by [3], and [22], across 60 epochs with a 0.1 aspect validation split.

Given that the collected data exhibit time-series characteristics, training with standard Recurrent Neural Network (RNN)-based models can lead to challenges such as vanishing gradients and the degradation of signal information over prolonged backpropagation sequences. To mitigate these issues, Long Short-Term Memory (LSTM) networks are frequently utilized due to their ability to preserve pertinent signal features across extended time steps within the hidden layers of complex RNN architectures. In the proposed model, the initial input stream processes pre-processed signal data, employing LSTM layers to address vanishing gradient concerns, with ReLU activation functions applied for outcome transformations.

4.3 Summary

This chapter concludes by summarizing our exploration of the dataset format and the behavior of 6DoF sensor-based data. This type of data can be effectively analyzed within the frequency domain using mathematical extraction techniques. However, it still necessitates feature selection, data preprocessing, and feature extraction before the dataset can serve as a foundation for model construction.

We observed and analyzed the data in the frequency domain using a Periodogram approach, which revealed motor patterns associated with PD symptoms. Furthermore, the proposed architecture was presented, primarily focused on two baselines and the interpretation of Fourier representations within the model's learning curve. We addressed concerns regarding data vanishing due to long-period time-series and suggestions for a double baseline with subnet feeding. These challenges were addressed by adhering to established research principles and incorporating the suggested baseline and data handling methods to achieve optimal performance.

Chapter 5 Result and Discussion

In this chapter, the findings derived from previous chapters are exhibited and discussed. The data distribution analysis demonstrates the methodology employed for attribute consideration, while the window selection process proves the impact of window sizing on model performance. Our primary focus, frequency analysis, is highlighted through the Periodogram approach, which reveals patterns for understanding finger motor characteristics across PD stages. Lastly, the constructed models are validated against a baseline model, with a comparative discussion presented to contextualize our work, including performance comparisons and cost analysis. This chapter, therefore, serves as the conclusion of our research.

5.1 Data Distribution

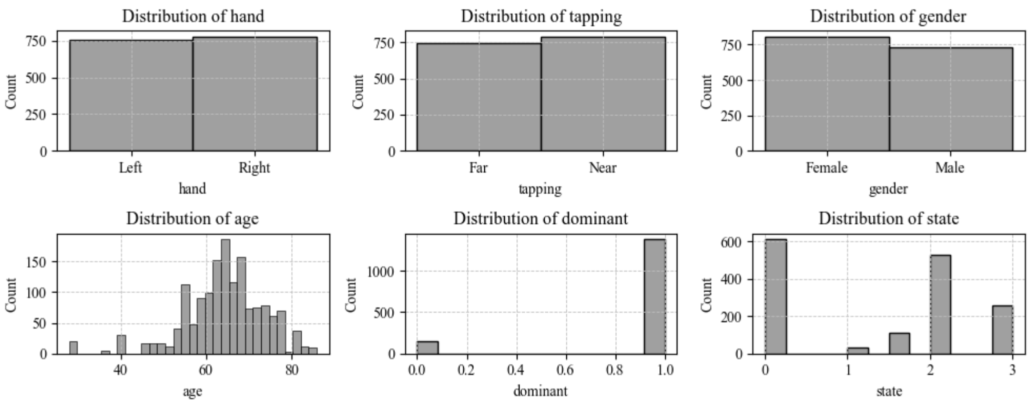

Figure 10 Histograms of record condition features illustrate the data distribution.

This section presents the distribution of each attribute remaining in the record conditions, as depicted in Figure 10. Features such as "hand," "tapping," "gender," "age," "dominant," and "PD stage" were plotted to examine their scattering. For attributes like "hand" and "gender," which contain only two possible options (e.g., left-hand or right-hand), the distributions are represented by two-bar graphs. These graphs primarily illustrate the comparative data weight between these two values.

Observations indicate that these attributes exhibit a relatively balanced distribution between their respective values.

Similarly, the "tapping" attribute, derived from the keystroke-tapping task, also presents only two options: "near" and "far" tapping. Consequently, its distribution aligns with the balanced characteristics observed for "hand" and "gender." Therefore, it is concluded that "hand," "tapping," and "gender" are suitable for inclusion in the model construction phase due to their balanced distributions.

Asymmetric distributions are presented for the "age," "dominant," and "PD stage" attributes. While the "dominant" attribute similar to previously mentioned features, offers only two values, its plot illustrates a significant imbalance with heavy skew towards the use of the dominant hand during the keystroke-tapping test. This indicates that the majority of patients predominantly use their dominant hand by default. Consequently, this column can be pruned to reduce dimensionality. For the "age" attribute, the plot exhibits a skewed distribution, with the mode falling within the 60 to 70 age range and there are some datapoint between 20 to 50. This range is typically when the disease onset occurs while the younger existing may indicate the control (non-PD) class. This issue can be mitigated by applying standardization, shaping center of the plot to approximately 70.

The "PD stage" attribute presents a different scenario. Within the model construction pipeline, this serves as the supervised label for the classification model. Although the plot exhibits a skewed distribution and a notable lack of data for PD stage 1, the available preprocessing methods are limited. In the graph, we observe the presence of stage 1.5 alongside stage 1. Medical opinion suggests that early-stage symptoms typically manifest on one side of the body, therefore, the label "1.5" indicates PD level 1 with symptoms affecting half of the body. In conclusion, we concatenate PD stages 1 and 1.5 to represent a single "level 1 PD" for our model's labeling. However, the imbalance in this attribute persists, which could affect overall performance. This issue may be mitigated by employing stratified sampling.

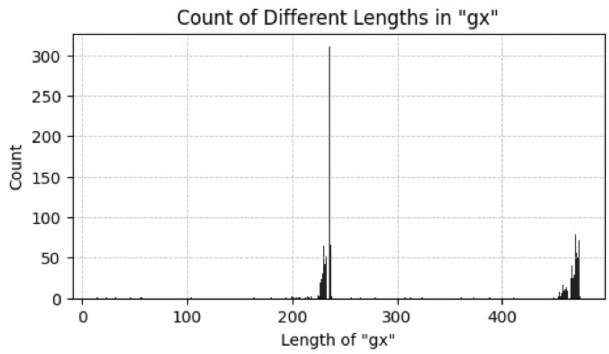

Figure 11 Signal length distribution plot

Additionally, Figure 11 illustrates the collected signal lengths from the keystroke-tapping task. The majority of signal lengths cluster around approximately 230 samples, indicating that the sensor was able to capture roughly 200 data points accumulatively within a 15-second period.

5.2 Windowing Selection

The window sizing selection is applied to observe an optimal period for the analysis procedure. To clarify this, the frequency analysis may affect from a long-time duration when we apply the estimator to visualize the pattern. This can cause lower precision while the period is longer [10]. Therefore, the optimize time duration was considered to prevent overfitting in the model. We created a sample set of window sizing including size 100 data points, 150 data points, 200 data points, 250 data points, 300 data points, and 350 data points. Then all the windowing sizes were applied to the signal data, then, running the pipeline to construct a binary classification model.

In Figure 12, the plot illustrates the accuracy of models constructed with varying signal window sizes. The trend line indicates that changes in window size do not significantly impact on the overall model performance. However, a window size of 250 was chosen as the standard for this study. This decision was primarily driven by the fact that the majority of signal lengths approximately reached 230, closely matching the 250-sample window. This selection ensures that the largest possible window size is utilized, thereby maximizing the information considered during model training without significantly interfering with overall accuracy.

Window size	Accuracy (%)
100	57.79
150	59.74
200	57.14
250	56.82
300	61.04
350	58.12

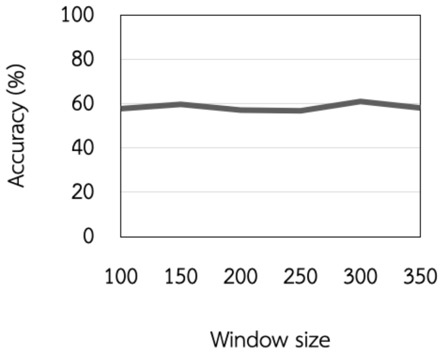

Figure 12 Illustrates the accuracy achieved by each model, corresponding to varying signal window sizes. The x-axis represents the window size, while the y-axis displays the resulting accuracy.

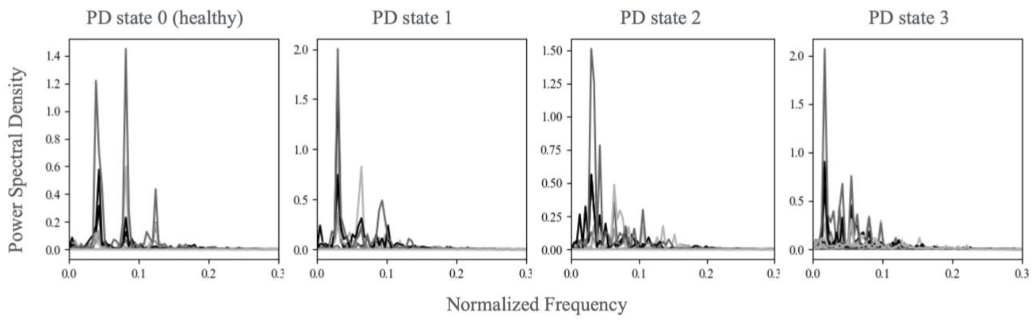

Figure 13 Frequency domain plot of the normalized 6DoF signal.

5.3 Periodogram Analysis

The normalized signal underwent Fast Fourier Transform (FFT) and was subsequently visualized in the frequency domain via a periodogram plot. The Y-axis represents signal power or the power spectral density (PSD), while the X-axis displays the frequency on a normalized scale.

As shown in Figure 13, the healthy group exhibits two prominent high PSD peaks between a normalized frequency of 0.0 and 0.1, accompanied by a moderate peak near the 0.1 mark. As the disease progresses to PD stage 1, these prominent peaks begin to shift toward the lower frequency area (to the left side). We observe that the high PSD peaks merge, resulting in only one high PSD peak near 0.0, with two moderate PSD peaks remaining below the 0.1 normalized frequency.

As the disease progresses to stage 2, a high PSD peak persists within the low-frequency range. However, the moderate peaks exhibit fluctuation and continue to shift towards the lower frequency band. These distinct moderate peaks become overlaid with the noisy baseline in the low-frequency area. A similar behavior is observed in PD stage 3, characterized by only one peak at low frequency accompanied by high-volume noise.

To discuss this behavior, while patient appendages begin to exhibit dysfunction, expressing tremors or stiffness in the hands, arms, or legs. Patients may demonstrate significant shaking due to muscle rigidity, which typically produces low-amplitude spectral signals across mid-to-high frequencies. For instance, during a keystroke task, finger movements might generate a low-to-mid frequency signal. In advanced stages of PD, tremors may lead to uncertain movements characterized by low amplitude power with a wider range of higher frequencies as background movement. This phenomenon could explain why PD stages 2 and 3 exhibit fluctuating higher noise levels towards the higher frequency areas in the Periodogram plot.

Conversely, a healthy group displays a cleaner signal peak with minimal background noise at higher frequencies.

5.4 Model Performance Evaluation

Our study introduces four distinct classification models, each evaluated using confusion metrics and classification reports.

Table 2 Model performance comparison

Model	Classification	Accuracy	Macro-sensitivity	FNR
Proposed binary-class model	Binary-class	78%	81%	23%
Proposed multi-class model	Multi-class	74%	74%	9%
Proposed ensemble model	Multi-class	60%	50%	41%
Proposed model with x-axis based	Multi-class	68%	62.5%	23%

The first model developed focuses on binary classification, distinguishing between healthy individuals and PD patients. From this, PD stages 1, 2, and 3 were concatenated into a single "PD class," then paired with the healthy group for model training. As a result, the dataset comprised two primary label classes: PD patient and non-PD (healthy). As detailed in Table 2, this model achieved an overall accuracy of 78%. It also demonstrated a false-negative rate (FNR) of 23%, with sensitivity of 77% for the healthy class and 78% for the PD class.

The second model performs multi-class classification specifically for PD severity levels 1, 2, and 3. For this model, the healthy control group was excluded, and only patient data were utilized for training. This approach was intended to create a model capable of classifying patients based solely on their PD stage. This model yielded accuracy of 74%, 9.7% FNR, and 74% macro-sensitivity

The third model integrates the two previous models into an ensemble framework designed to predict non-PD (healthy), PD level 1, 2, and 3. This approach accomplished 60% overall accuracy and a 23.5% FNR. The sensitivities for the respective classes were 81% (non-PD), 25% (PD level 1), 55% (PD level 2), and 40% (PD level 3). These outcomes undergo the expected performance and suggest that the ensemble approach may not be the most effective strategy for this classification task.

The final model exclusively uses gyroscope X-axis signals from both channels as standalone input features. This approach was inspired by the seasonal plot results, detailed in the appendix. Specifically, seasonal plots with window sizes of 15 and 25 revealed distinct patterns in the "gx" (gyroscope x-axis) data from the middle finger. For the healthy group, this plot exhibited a sine-wave pattern, whereas PD stage 3 displayed a linear trend. These observed differences prompted us to test this model to confirm the relationship of these patterns. This model achieved 68% accuracy and a 14.5% false-negative rate (FNR). Its sensitivities were 77% for non-PD, 50% for PD level 1, 67% for PD level 2, and 56% for PD level 3. While this performance indicates a slight improvement over the ensemble model, the differentiation in patterns may not have significantly impacted the overall model performance.

Macro-sensitivity, as illustrated in Table 2, provides a summary average of the model's sensitivity. For instance, the multi-class model's macro-sensitivity represents the average sensitivity across PD stages 1, 2, and 3, even though individual class sensitivities may not precisely match this mean value. To analyze each class's performance individually, the confusion matrix and classification report offer more precise insights and clarification through their True/False Positive/Negative values and micro-sensitivity value, respectively. The confusion matrix illustrates in Figure 14 and the classification report are presented in Table 3.

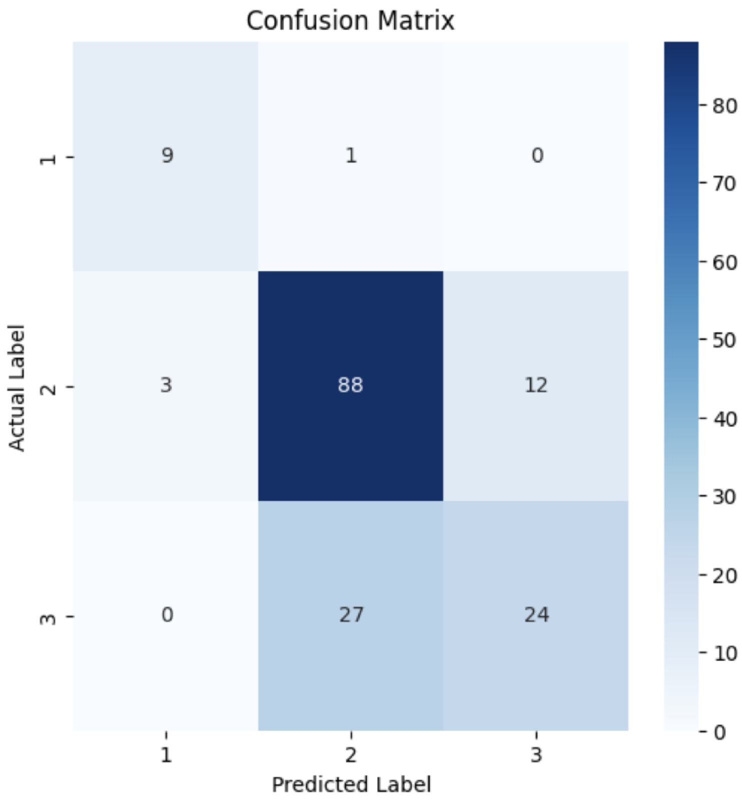

Figure 14 Proposed multi-class model confusion matrix

Table 3 Proposed second multi-class model classification report

Class	Precision	micro-Sensitivity	F1-Score
PD stage 1	75%	90%	82%
PD stage 2	76%	85%	80%
PD stage 3	67%	47%	55%

Despite the earlier observations, the classification report from the second model reveals significant results. As detailed in Table 3, Class 1 PD achieved a micro-sensitivity of 90%, while Class 2 and Class 3 reached 85% and 47% sensitivity, respectively. This performance indicates and supports the model's potential for early-stage PD classifying within the keystroke-tapping approach and frequency domain analysis.

5.5 Baseline Comparison and Discussion

Table 4 illustrates the comparison between our proposed model and baseline models. The comparison result indicates that our proposed models perform slightly below the baseline. However, an examination of each individual class performance, especially micro-sensitivity as presented in Table 3, reveals a significant strength. Our multi-class model achieves 90% sensitivity for early-stage (class 1) detection with less than 10% FNR. This high sensitivity for early-stage detection, combined with the accessibility of the clinical test in terms of cost-effectiveness, time efficiency, and minimal resource requirements, underscores the primary advantages of our approach.

Table 4 Baseline performance comparison

Model	Classification	Accuracy	Macro-sensitivity	FNR
Proposed binary-class model	Binary-class	78%	81%	23%
Proposed multi-class model	Multi-class	74%	74%	9%
Hathaliya et al. [21]	Binary-class	92.4%	N/A	N/A
B. Vidya and Sasikumar P. [22]	Multi-class	98.32%	98.29%	N/A

Due to the significant imbalance in our dataset, the proposed model might struggle to effectively learn and generalize, potentially leading to overfitting or underfitting in under-represented cases. To tackle this, feeding additional data may help model generalization and performance development.

Unlike the method proposed by [22], which depends on a VGRF dataset that requires specialized equipment and controlled environment, our signal collection technique is more accessible and cost-effective. While similar to the approach in [21], our method supports multi-class classification with acceptable sensitivity for detecting PD level 1. This makes our approach a promising tool for early-stage Parkinson's disease detection under limited resources.

5.6 Cost Analysis

This section provides a cost analysis to illustrate the trade-off between the model's performance and the accessibility of disease screening. The proposed keystroke-tapping task requires two MPU6050 modules, which are attached to the patient's fingers. Based on Q2 2025 market prices in Thailand, each module costs approximately 100 THB to 200 THB, and the total implementation cost, including other necessary components, is not expected to exceed 1,000 THB.

In contrast, the approach described in [22] necessitates more extensive equipment, including motion cameras for trajectory capture, an inertial measurement unit (IMU) device, pressure sensors for step pressure readings, and force-sensitive resistors (FSRs) for signal reception. While the IMU device's price is comparable to the MPU6050 module, individual FSRs can cost up to approximately 300 THB. Excluding the cameras, a full implementation requiring 16 FSRs could escalate the total cost to around 5,000 THB, particularly when factoring in the need for a controlled environment for camera capturing during the task.

Considering these prerequisites and the respective performance levels, our proposed method offers greater cost-efficiency, pre-screening availability, and enhanced flexibility during the data collection procedure. The trade-off in performance, however, warrants careful consideration in the overall assessment.

5.7 Limitation

A primary limitation of this study lies in its reliance on finger-based symptom capture, which may not fully encompass the diverse presentation of PD symptoms, particularly those not indicated by impaired finger dexterity. Consequently, finger movement dysfunction might not be an ideal early-stage indicator if a patient's initial symptoms manifest in other body parts. However, this approach proves beneficial when the affected area aligns with the finger monitoring procedure and within contexts of resource limitation. Conversely, as patients progress to higher PD levels, they often experience walking or standing disabilities. In such cases, finger monitoring can still be useful by revealing PD patterns within the frequency domain that correlate with overall disease progression.

5.8 Summary

This chapter has provided an understanding of data distribution, including sample imbalance issues, considerations regarding the appropriate data period for model construction, as well as pattern inspection under the frequency domain. This can be summarized by the development of four distinct models, which the multi-class model stands out for its remarkable capability in early-stage detection. While the overall performance may not fully compete with baseline models, our approach offers an alternative of more accessible and cost-effective.

Chapter 6 Conclusion

This chapter provides an overview of our work, beginning with a recap of our model's conception and foundational ideas. Subsequently, our contributions will be presented to highlight what has been delivered. Finally, potential future work aimed at improving our model will be discussed.

6.1 Thesis Summary

Developing accurate and accessible tools for early-stage PD screening and assessing its severity is essential for improving patient outcomes. Although many approaches have been suggested, they often encounter limitations concerning cost, practicality, or their ability to detect subtle motor impairments. This study introduces a novel method to overcome these challenges by integrating a simple, low-cost keyboard-tapping task with advanced signal processing and machine learning techniques.

By using data from a 6DoF sensor and leveraging frequency-domain analysis with LSTM networks, our model shows promising results in classifying PD severity levels 1-3. Specifically, the model reached an overall accuracy of 74%, with encouraging sensitivities of 90% for level 1, 85% for level 2, and 47% for level 3. We recognize that achieving high accuracy in early-stage detection is still a significant challenge. However, these findings underscore the potential of our method to improve screening, especially in areas with limited access to specialized testing.

6.2 Contributions

This thesis explores several avenues for developing more accessible and cost-effective methods with competitive sensitivity for early-stage PD detection.

- We implemented a novel multi-class classification model trained on sensor data. This model is capable of classifying PD levels 1, 2, and 3 based on the Hoehn and Yahr scale [19].
- The model demonstrated a competitive sensitivity with 90% recall for PD level 1 and an overall accuracy of 74%, utilizing a dataset derived from a keystroke-tapping task.
- This approach offers an alternative PD screening method that provides acceptable accuracy, enhanced accessibility, and greater cost-efficiency compared to existing alternatives.

6.3 Future Work

Future research will concentrate on enhancing the model's accuracy and generalization capabilities, particularly since the current model's performance slightly trails the baseline due to the time and resource limitations encountered during this research. Future efforts will involve incorporating larger, more diverse datasets, thoroughly re-investigating preprocessing techniques, and exploring alternative deep learning architectures. For example, attributes like sex and hand, which contain only two unique values, may not have been optimally aligned with the multi-class classification labels in the attempted experiment, potentially not directly reflecting their impact on overall performance. Furthermore, the criteria for window size selection may not have fully aligned with the sensor's sampling rate. Specifically, the majority of signal lengths contain approximately 230 data points within a 15-second sampling period, suggesting a sampling rate of around 15 samples/second. However, our experiments utilized a window size of 50 data points, which might not be congruent with this underlying sampling rate. This potential misalignment could explain why variations in window sizing did not significantly affect model performance.

Therefore, future work on window sizing experiments will require validation to ensure alignment with the sensor's inherent sampling rate.

The curse of dimensionality is a primary contributor to model overfitting. One potential optimization to reduce dimensionality involves applying frequency binning to simplify data representation. For instance, instead of feeding a long period signal directly, frequencies could be grouped into defined frequency range bins, where the count within each bin represents the spectral density of that frequency range. This approach may simplify dataset complexity and dimensionality, potentially improving overall research performance. Ultimately, the proposed model should be capable of classifying healthy individuals as well as PD levels 1, 2, and 3. This full-loop classification within our proposed model would enable a stronger comparison and better alignment with baseline models.

Additionally, integrating clinical data, such as patient demographics and medical history, could further improve its predictive power. Ultimately, by advancing the development of accurate, accessible, and non-invasive tools for early Parkinson's Disease detection and assessment, this research aims to contribute to a future where individuals receive timely interventions, thereby slowing disease progression and preserving their quality of life.

6.4 Summary

In time-series domain, this thesis offers a step-by-step analysis within the time-series domain, encompassing data distribution, Periodogram analysis, and the Fourier learning model. We've also suggested several future works to enhance the overall model performance. Our aim is for this research to serve as a foundational resource for clinicians seeking to develop more accessible methods to assist patients.

REFERENCES

[1] G. E. Hearn and A. V. Metcalfe, “5 - Frequency representation of random signals,” in Spectral Analysis in Engineering, G. E. Hearn and A. V. Metcalfe, Eds., Oxford: Butterworth-Heinemann, 1995, pp. 109–142. doi: 10.1016/B978-034063171-3/50007-5.

[2] S. Nuvoli et al., “[123I]Metaiodobenzylguanidine (MIBG) Cardiac Scintigraphy and Automated Classification Techniques in Parkinsonian Disorders,” Molecular Imaging and Biology, vol. 22, no. 3, pp. 703–710, Jun. 2020, doi: 10.1007/s11307-019-01406-6.

[3] R. Ahmed Bhuiyan, N. Ahmed, M. Amiruzzaman, and M. R. Islam, “A Robust Feature Extraction Model for Human Activity Characterization Using 3-Axis Accelerometer and Gyroscope Data.,” Sensors (Basel), vol. 20, no. 23, Dec. 2020, doi: 10.3390/s20236990.

[4] M. Belić, V. Bobić, M. Badža, N. Šolaja, M. Đurić-Jovičić, and V. S. Kostić, “Artificial intelligence for assisting diagnostics and assessment of Parkinson’s disease—A review,” Clinical Neurology and Neurosurgery, vol. 184, p. 105442, Sep. 2019, doi: 10.1016/j.clineuro.2019.105442.

[5] F. Segovia, J. M. Górriz, J. Ramírez, F. J. Martínez-Murcia, and D. Castillo-Barnes, “Assisted Diagnosis of Parkinsonism Based on the Striatal Morphology,” Int. J. Neur. Syst., vol. 29, no. 09, p. 1950011, Nov. 2019, doi: 10.1142/S0129065719500114.

[6] A. Subasi, “Chapter 2 - Data preprocessing,” in Practical Machine Learning for Data Analysis Using Python, A. Subasi, Ed., Academic Press, 2020, pp. 27–89. doi: 10.1016/B978-0-12-821379-7.00002-3.

[7] A. Meyer-Baese and V. Schmid, “Chapter 2 - Feature Selection and Extraction,” in Pattern Recognition and Signal Analysis in Medical Imaging (Second Edition), A. Meyer-Baese and V. Schmid, Eds., Oxford: Academic Press, 2014, pp. 21–69. doi: 10.1016/B978-0-12-409545-8.00002-9.

[8] J. M. Blackledget, “Chapter 4 - The Fourier Transform,” in Digital Signal Processing (Second Edition), J. M. Blackledget, Ed., Woodhead Publishing, 2006, pp. 75–113. doi: 10.1533/9780857099457.1.75.

[9] A. F. Siegel and M. R. Wagner, “Chapter 14 - Time Series: Understanding Changes Over Time,” in Practical Business Statistics (Eighth Edition), A. F. Siegel and M. R. Wagner, Eds., Academic Press, 2022, pp. 445–482. doi: 10.1016/B978-0-12-820025-4.00014-2.

[10] J. DEMPSTER, “CHAPTER SIX - Signal Analysis and Measurement,” in The Laboratory Computer, J. DEMPSTER, Ed., London: Academic Press, 2001, pp. 136–171. doi: 10.1016/B978-012209551-1/50039-8.

[11] F. Aksan, Y. Li, V. Suresh, and P. Janik, “CNN-LSTM vs. LSTM-CNN to Predict Power Flow Direction: A Case Study of the High-Voltage Subnet of Northeast Germany.,” Sensors (Basel), vol. 23, no. 2, Jan. 2023, doi: 10.3390/s23020901.

[12] G. M. Halliday, K. Del Tredici, and H. Braak, “Critical appraisal of brain pathology staging related to presymptomatic and symptomatic cases of sporadic Parkinson’s disease,” in Parkinson’s Disease and Related Disorders, P. Riederer, H. Reichmann, M. B. H. Youdim, and M. Gerlach, Eds., Vienna: Springer Vienna, 2006, pp. 99–103.

[13] P. Maiti, J. Manna, and G. L. Dunbar, “Current understanding of the molecular mechanisms in Parkinson’s disease: Targets for potential treatments.,” Transl Neurodegener, vol. 6, p. 28, 2017, doi: 10.1186/s40035-017-0099-z.

[14] P. Panyakaew et al., “Exploring the Complex Phenotypes of Impaired Finger Dexterity in Mild-To-moderate Stage Parkinson’s Disease: A Time-Series Analysis,” Journal of Parkinson’s Disease, vol. 13, no. 6, pp. 977–990, 2023, doi: 10.3233/JPD-230029.

[15] P. F. Bernath, “Fourier Transform Techniques☆,” in Encyclopedia of Analytical Science (Third Edition), P. Worsfold, C. Poole, A. Townshend, and M. Miró, Eds., Oxford: Academic Press, 2019, pp. 81–89. doi: 10.1016/B978-0-12-409547-2.14518-4.

[16] C. R. Pereira et al., “Handwritten dynamics assessment through convolutional neural networks: An application to Parkinson’s disease identification,” Artificial Intelligence in Medicine, vol. 87, pp. 67–77, May 2018, doi: 10.1016/j.artmed.2018.04.001.

[17] M. Webber and R. F. Rojas, “Human Activity Recognition With Accelerometer and Gyroscope: A Data Fusion Approach,” IEEE Sensors Journal, vol. 21, no. 15, pp. 16979–16989, Aug. 2021, doi: 10.1109/JSEN.2021.3079883.

[18] J. Mei, C. Desrosiers, and J. Frasnelli, “Machine Learning for the Diagnosis of Parkinson’s Disease: A Review of Literature,” Frontiers in Aging Neuroscience, vol. 13, 2021, [Online]. Available: https://www.frontiersin.org/journals/aging-neuroscience/articles/10.3389/fnagi.2021.633752

[19] C. G. Goetz et al., “Movement Disorder Society Task Force report on the Hoehn and Yahr staging scale: status and recommendations.,” Mov Disord, vol. 19, no. 9, pp. 1020–1028, Sep. 2004, doi: 10.1002/mds.20213.

[20] Z. Wang et al., “Multi-modal classification of neurodegenerative disease by progressive graph-based transductive learning,” Medical Image Analysis, vol. 39, pp. 218–230, Jul. 2017, doi: 10.1016/j.media.2017.05.003.

[21] J. J. Hathaliya, H. Modi, R. Gupta, S. Tanwar, P. Sharma, and R. Sharma, “Parkinson and essential tremor classification to identify the patient’s risk based on tremor severity,” Computers and Electrical Engineering, vol. 101, p. 107946, Jul. 2022, doi: 10.1016/j.compeleceng.2022.107946.

[22] B. Vidya and Sasikumar P., “Parkinson’s disease diagnosis and stage prediction based on gait signal analysis using EMD and CNN–LSTM network,” Engineering Applications of Artificial Intelligence, vol. 114, p. 105099, Sep. 2022, doi: 10.1016/j.engappai.2022.105099.

[23] J. Jankovic, “Parkinson’s disease: clinical features and diagnosis.,” J Neurol Neurosurg Psychiatry, vol. 79, no. 4, pp. 368–376, Apr. 2008, doi: 10.1136/jnnp.2007.131045.

[24] H. Tsutsumi, K. Kondo, K. Takenaka, and T. Hasegawa, “Sensor-Based Activity Recognition Using Frequency Band Enhancement Filters and Model Ensembles,” Sensors, vol. 23, no. 3, 2023, doi: 10.3390/s23031465.

[25] C. Váradi, K. Nehéz, O. Hornyák, B. Viskolcz, and J. Bones, “Serum N-Glycosylation in Parkinson’s Disease: A Novel Approach for Potential Alterations,” Molecules, vol. 24, no. 12, 2019, doi: 10.3390/molecules24122220.

[26] H. Ma, T. Tan, H. Zhou, and T. Gao, “Support Vector Machine-recursive feature elimination for the diagnosis of Parkinson disease based on speech analysis,” in 2016 Seventh International Conference on Intelligent Control and Information Processing (ICICIP), Dec. 2016, pp. 34–40. doi: 10.1109/ICICIP.2016.7885912.

[27] T. D. Pham and H. Yan, “Tensor Decomposition of Gait Dynamics in Parkinson’s Disease,” IEEE Transactions on Biomedical Engineering, vol. 65, no. 8, pp. 1820–1827, Aug. 2018, doi: 10.1109/TBME.2017.2779884.

[28] M. A. Farahani, M. R. McCormick, R. Harik, and T. Wuest, "Time-series classification in smart manufacturing systems: An experimental evaluation of state-of-the-art machine learning algorithms," Robotics and Computer-Integrated Manufacturing, vol. 91, p. 102839, Feb. 2025, doi: 10.1016/j.rcim.2024.102839.

APPENDIX

APPENDIX A: Seasonal Plot

In our data exploration, the seasonal plot was utilized to observe the inherent seasonal behavior within the time-series data. A seasonal plot, also recognized as time series with duration analysis, serves to visualize time-series data over a specific period [9]. Its primary focus relies on the seasonal component, which indicates effects occurring within a defined period. For example, the weather forecasts across seasons within a year. This methodology was applied to our time-series data by varying the duration size (window size) and implementing it across our dataset.

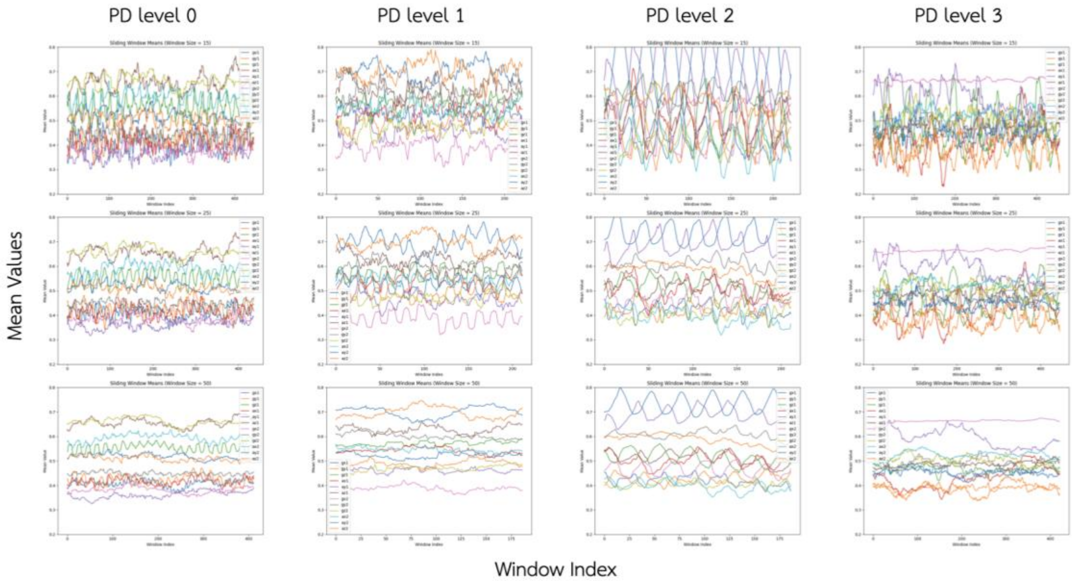

Figure 15 Seasonal plot with varying window size

Window sizes are applied to sample data, and the data points within each window are then calculated as the average. This window then slides to a new starting point, and the calculation process repeats. In essence, applying the seasonal plot to this dataset can be considered as a moving average plot.

As illustrated in Figure 14, the first row of each column displays a window size of 15, meaning the mean value of 15 sampled data points is calculated. The second and third rows correspond to window sizes of 25 and 50, respectively. These samples were obtained from a patient performing the "far" tapping task with their left hand. This process plotted as a seasonal signal to observe recurring patterns.

In Figure 14, let’s focusing on the pink-color "gx2" instance at a window size of 25, distinct patterns emerge across PD stages. Specifically, PD Stage 0 exhibits a complex sine wave, while PD Stage 1 displays a smoother sine wave. For PD Stage 2, there is a slight upward in its mean, with the overall trend remaining a sine wave. Notably, the mean for PD Stage 3 sharply increases from the 0.4-0.5 range to the 0.6-0.7 range, accompanied by signs of a linear trend.

In this discussion, the progression of Parkinson's Disease (PD) appears to be related to "far" tapping behavior. When a patient performs "far" tapping with their left hand, the middle finger strikes "q" while the index finger should strike "]". This observation suggests that the development of PD may lead to hand tremors specifically affecting the index finger, given that the "gx2" sensor is attached to this digit. However, this assumption requires further empirical evidence and experimentation for conclusive proof.

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